| 5th Semester | | |
|---|---|---|
| **ISCED Code** | **Course Code** | **Course Title** |
| **0714** | **EEE-2421** | **Electrical Drives and Instrumentation** |
| **Credit Hours:** 2 | **Contact Hours:** 2 | **Type:** Core Courses |
| **Prerequisite: EEE-1221 (Electronics)** | | |
| **Co-requisite:EEE-2422 (Electrical Drives and Instrumentation Lab** | | |

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|---|---|---|---|
| **Course Assessments** | **CIE**: Continuous Internal Evaluation | Attendance | 10 Marks |
| | | Class test/ Assignment/ Quizzes | 10 Marks |
| | | Mid-term | 30 Marks |
| | **SEE**: Semester End Examination | | 50 Marks |
| | **Total** | | 100 Marks |

**Course Rationale:**

This course aims to equip students with the knowledge of transformer, DC and AC electrical drive systems as well as DC/AC motors and generators which are commonly used in industries. Students will learn about synchronous and induction motors, essential for designing and controlling various devices and applications. This knowledge helps improve the performance and efficiency of systems that rely on these electrical machines. The course includes practical projects based on measurement and sensors that teach industry-relevant skills, preparing students for careers where integrating hardware and software is crucial.

**Objectives:**

The material will be presented using the normal mix of lectures and laboratory experiments and demonstrations. This course objective is also to

1. Understand the principles and applications of DC motors, synchronous motors, and transformers.
2. Master measurement and instrumentation techniques essential for electrical engineering practice.
3. Analyze the performance and characteristics of DC motors, synchronous motors, and transformers through practical experiments.
4. Apply theoretical knowledge to solve real-world problems in electrical machines and instrumentation.

**Course Learning Outcomes (CLOs):**

Upon successful completion of this course, students will be able to:

| # | Course Learning Outcomes | Weightage (%) |
|---|---|---|

| | | |
|---|---|---|
| 1. | **Explain** the knowledge of Electrical (Static and Rotating) Machines and their controls(Drives) | 40 |
| 2. | **Identify and Recognize** various measurement techniques for both electrical and non-electrical quantities. | 30 |
| 3. | **Analyze** different types of numerical problems regarding Electrical machineries and measurement devices. | 30 |

**Mapping of CLO-PLO-DL-KP-EP-EA:**

| # | CLOs | PLOs | DL | KP | EP | EA | Teaching Learning Strategy (TLS) | Assessment Strategy (AS) |
|---|---|---|---|---|---|---|---|---|
| CLO1 | **Explain** the knowledge of Electrical (Static and Rotating) Machines and their controls (Drives) | PLO1 | K1 | C2 | - | - | Lecture, Class discussion, Assignment, Note | Exam, Quiz, Assignment |
| CLO2 | **Identify and Recognize** various measurement techniques for both electrical and non-electrical quantities. | PLO2 | K3 | C4 | - | - | Lecture, Class discussion, Assignment, Problem solving session | Assignment, Class Test, Exam |
| CLO3 | **Analyze** different types of numerical problems regarding Electrical machineries and measurement devices. | PLO2 | K3 | C2 | - | - | Lecture, Class discussion, Assignment, Note, | Assignment, Class Test, Exam, Presentation |

**Course Content:**

| # | Content | Duration | CLOs |
|---|---|---|---|
| | Mid-Term (30 Marks) | | |
| 1 | Introduction to Electrical Drives: Introduction to electrical | 2 weeks | CLO1,CLO3 |

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|---|---|---|---|
| | machines. Rotational motion, Newton's law, and power relationships. Magnetic field, Faraday’s law, induced voltage on a conductor moving in a magnetic field, production of force on a wire in a magnetic field, Classification of Load torques, Mechanical and Electrical Power Calculation, Sizing of electric motors for given load system, Classes of Motor Duty, De-rating factor for electric motor sizing, Energy Efficient Motors, Motor nameplate. | | |
| 2 | Transformers: Single-phase transformers: Construction, principle of operation and equivalent circuit, phasor diagram, efficiency and regulation. Short and open circuit tests. Three-phase transformers: Construction and connections. | 2 weeks | CLO1,CLO3 |
| 3 | DC Motor Drives: Principles of operation and construction of DC machines, Emf equation and principle of commutation. Controlled rectifier fed dc drives, Power factor, supply harmonics and ripple in motor current, Chopper controlled dc drives, Closed loop control of DC Drives, Two and four quadrant controls. | 2 weeks | CLO1,CLO3 |
| Section-B (Final Exam: 50 Marks)<br>Group-A (20 Marks) | | | |
| 4 | Induction Motor Drives: Principles of operation and basic construction of three-phase induction motor. Slip equation, equivalent circuit, determination of equivalent circuit parameters by no-load and blocked-rotor tests. Volts per Hertz drives, Flux vector control drives, Direct torque control drives, Soft-Starters, Selection of speed drives and soft starters, Line reactors, Troubleshooting of AC Drives system, Drives parameter programming. | 2 weeks | CLO1,CLO3 |
| 5 | Synchronous Motor Drives: Stepper Motor Drives, Switched reluctance motor drives, Permanent Magnet Synchronous Motor (Interior Permanent Magnet and Surface Permanent Magnet, Brushless DC motor). Thyristor and microprocessor-based speed control of motors. | 2 weeks | CLO1,CLO3 |
| Group-B (30 Marks) | | | |
| 6 | General Concepts of Measurements and instrumentations: Variables and measurement signals, the three stages of generalized measurement system, some common terms used in | 2 weeks | CLO1,CLO2 |

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| | the measurement system, mechanical loading, impedance matching, and frequency response. Factors considered in selection of instruments - Measurement accuracy and precision. Error analysis and classification, sources of error. | | |
| 7 | Instrumentation amplifiers: Differential, logarithmic and chopper amplifiers; Frequency and voltage measurements using digital techniques; Recorders and display devices, spectrum analyzers and logic analyzers | 1 week | CLO1,CLO2 |
| 8 | Measurement of Non Electrical Quantities: Transducers-terminology, types, principles and application of photovoltaic, piezoelectric, thermoelectric, variable reactance and optoelectronic transducers; Measurement of Temperature: Resistance thermometer, and thermo couples. | 2 weeks | CLO1,CLO2 |

**Marks Distribution:**

**Course Assessment Pattern (Theory courses):**

| Bloom's Category | | Evaluations out of 100 marks | | | |
|---|---|---|---|---|---|
| | | CIE (50 marks) | | | SEE (50marks) |
| Cognitive learning | Affective learning | Mid-term (30) | Assignment/Class Test (10) | Attendance Marks (10) | Written Exam (50) |
| Remember | | | | - | 5 |
| Understand | 5 | 5 | | - | 15 |
| Apply | 10 | - | | - | 20 |
| Analyze | 15 | 5 | | - | 10 |
| Evaluation | - | | | - | |
| Create | | - | | - | |
| | Responding | x | x | 10 | |
| Total allocated marks | | 30 | 10 | 10 | 50 |

**Grading Policy:** As per IIUC grading policy

**Learning Materials:**

**Text Books:**

| Name of Authors | Title of Book | Edition | Publisher's Name | Year | ISBN |
|---|---|---|---|---|---|
| Malaric, R. | Instrumentation and Measurement in Electrical Engineering | N/A | Brown Walker Press | 2011 | N/A |
| Stephen J. | Electric Machinery Fundamental | N/A | N/A | N/A | N/A |

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|---|---|---|---|---|---|
| Chapman | | | | | |
| B.L. Theraja & A.K. Theraja | A textbook of Electrical Technology | N/A | N/A | N/A | N/A |

**Reference Books:**

| Name of Authors | Title of Book | Edition | Publisher's Name | Year | ISBN |
|---|---|---|---|---|---|
| A.K. Sawhney | A Course in Electrical and Electronic Measurement and Instrumentation | N/A | Dhanpat Rai & Sons, New Delhi | 2001 | N/A |
| Ernest O. Doeblin | Measurement Systems: Applications and design | N/A | Mc-Graw Hill | 2004 | N/A |
| T.G. Beckwith & N.L. Buck | Mechanical Measurements | N/A | Addition Wesley | 2007 | N/A |
| E.W. Golding & F.C. Widdis | Electrical Measurements and Measuring Instruments | N/A | A H Wheeler & Company, Calcutta | 1993 | N/A |

| **5th Semester** | | |
|---|---|---|
| **ISCED Code** | **Course Code** | **Course Title** |
| **0714** | **EEE-2422** | **Electrical Drives and Instrumentation Lab** |
| **Credit Hours: 1** | **Contact Hours: 2** | **Type:** Core Courses |
| **Prerequisite: EEE-1222 (Electronics Lab)** | | |
| **Co-requisite: EEE-2421 (Electrical Drives and Instrumentation)** | | |

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| Course Assessments | **CIE**: Continuous Internal Evaluation | Attendance | 10 Marks |
| | | Class test/ Assignment/ Quizzes | 10 Marks |
| | | Mid-term | 30 Marks |
| | **SEE**: Semester End Examination | | 50 Marks |

**Course Rationale:**

This course integrates Arduino learning and measurement to bridge theory and practice. Utilizing Arduino for hands-on projects enhances students' understanding of electrical drive systems through real-time data acquisition and control. This approach fosters practical skills and innovative problem-solving. The course prepares students to utilize modern tools in electrical engineering, enhancing their readiness for industry challenges.

**Objectives:**

The material will be presented using the normal mix of lectures and laboratory experiments and demonstrations. This course objective is also to

1. Provide practical experience with electrical machines, including motors and generators, through laboratory exercises.
2. Teach students to set up and program Arduino microcontrollers for controlling and

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| monitoring electrical motor drives, sensors etc.<br>3. Develop skills on how to use arduino for various purposes.<br>4. Equip students with the ability to use Arduino for real-time measurement and data acquisition of various parameters.<br>5. Facilitate the design and execution of lab projects that integrate Arduino with electrical machines, demonstrating the application of theoretical concepts in real-world scenarios.<br>6. Apply theoretical knowledge to solve real-world problems in electrical machines and instrumentation. |

**Course Learning Outcomes (CLOs):**

At the completion of the subject, students should be able to perform the following tasks:

| # | Course Learning Outcomes | Weightage (%) |
|---|---|---|
| **1.** | **Apply** engineering knowledge and follow instructions to perform experiments. | 40 |
| **2.** | **Perform** the experimental using appropriate engineering tools | 30 |
| **3.** | **Develop** a project based on achieved theoretical knowledge | 30 |

**Mapping of CLO-PLO-DL-KP-EP-EA:**

| # | CLOs | PLOs | DL | KP | EP | EA | Teaching Learning Strategy (TLS) | Assessment Strategy (AS) |
|---|---|---|---|---|---|---|---|---|
| CLO1 | **Apply** engineering knowledge and follow instructions to perform experiments. | PO1 | C3 | K3 | - | - | Lecture, Class discussion, Assignment, Note | Lab Report, Lab Performance, viva |
| CLO2 | **Perform** the experiments using appropriate engineering tools | PO5 | C3 | K6 | - | - | Lecture, Class discussion, Assignment, Note | Lab Performance, Lab Report, |
| CLO3 | **Develop** a project based on achieved theoretical knowledge | PO3 | C6 | K5 | - | - | Lecture, Class discussion, Assignment, Lab work, Note | Project, Report, viva |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | |
| **Note: DL:** Domain/level of learning taxonomy, **KP:** Knowledge Profile, **EP:** Attribute of Complex Engineering Problems, **EA:** Attribute of Complex Engineering Activities, **Learning Domains** (C: Cognitive, A: Affective, P: Psychomotor) | | | | | | | | |

**Course Content:**

| # | Content | Duration | CLOs |
|---|---|---|---|
| 1 | **Familiarization with instruments used in Electrical Drives Lab.** | **1 week** | **CLO1** |
| 2 | **Introduction to Microcontroller and Arduino uno R3:** Exploring the architecture, features, and programming of microcontrollers with a focus on Arduino Uno R3 for real-time control applications. | **1 week** | **CLO1** |
| 3 | **Arduino Uno R3 and beyond:** Utilizing Arduino Uno R3 to make powerful and impactful prototypes and going beyond for making innovations. | **2 weeks** | **CLO1, CLO2** |
| 4 | **Familiarization with different types of sensors:** Demonstrations of different sensors such as gas sensor, IR sensors and how they react to different projects. | **3 weeks** | **CLO2** |
| 5 | **Midterm Exam and Review**: Comprehensive review of previous experiments, viva voce to assess understanding, and a midterm evaluation. | **1 week** | |
| | ***MIDTERM*** | | |
| 6 | **Familiarization with output devices for Arduino Uno R3:** Understanding the functionality and integration of output devices like LEDs, buzzers, and displays with the Arduino Uno R3 for practical applications. | **3 weeks** | **CLO2** |
| 7 | **Learning to control with sensors:** Integrating sensors like gas sensor, IR, transducers with Arduino to enable real-time feedback and precise control of motor operations. | **2 weeks** | **CLO2** |
| 8 | **Explanation of Arduino-Based Projects:** To explain projects to students for giving them ideas for future projects. | **2 weeks** | **CLO3** |

**Marks Distribution:**

**Course Assessment Pattern (Lab courses):**

| | | **CIE (50 marks)** | | **SEE (50 marks)** |
|---|---|---|---|---|
| Cognitive learning | Affective learning | Attendance Marks (10) | Continuous Evaluation (40) | Final Exam / Project (50) |
| Remember | | | | 5 |
| Understand | | | 5 | 5 |
| Apply | | | 20 | 15 |
| Analyze | | | 5 | 10 |
| Evaluation | | | 5 | 10 |

| | | | | |
|---|---|---|---|---|
| Create | | | 5 | 5 |
| | Responding | 10 | | |
| **Total allocated marks** | | 10 | 40 | 50 |

**Grading Policy:** As per IIUC grading policy

**Learning Materials:**

**Text Books:**

| Name of Authors | Title of Book | Edition | Publisher’s Name | Year | ISBN |
|---|---|---|---|---|---|
| Malaric, R. | Instrumentation and Measurement in Electrical Engineering | N/A | Brown Walker Press | 2011 | N/A |
| Stephen J. Chapman | Electric Machinery Fundamentals | N/A | N/A | N/A | N/A |

**Reference Books:**

| Name of Authors | Title of Book | Edition | Publisher’s Name | Year | ISBN |
|---|---|---|---|---|---|
| A.K. Sawhney | A Course in Electrical and Electronic Measurement and Instrumentation | N/A | Dhanpat Rai & Sons, New Delhi | 2001 | N/A |
| Ernest O. Doeblin | Measurement Systems: Applications and design | N/A | Mc-Graw Hill | 2004 | N/A |
| T.G. Beckwith & N.L. Buck | Mechanical Measurements | N/A | Addition Wesley | 2007 | N/A |
| E.W. Golding & F.C. Widdis | Electrical Measurements and Measuring Instruments | N/A | A H Wheeler & Company, Calcutta | 1993 | N/A |
| B.L. Theraja & A.K. Theraja | A textbook of Electrical Technology | N/A | N/A | N/A | N/A |